Shanmukh Machiraju

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.83
FIITJEE SCHOOL	Class XII (Telangana Board of Intermediate Education)	2024	93.00%
FIITJEE SCHOOL	Class X (Telangana State Board of Secondary Education)	2022	9.20%

EXPERIENCE

Robotics and ML Intern at Rubotix.ai

- Built an autonomous navigation stack for a mecanum-wheeled robot as a modular **5-package ROS 2 Humble workspace** spanning description, simulation, navigation, hardware, and bringup, supporting holonomic autonomous and teleoperated control.
- Developed a Gazebo Sim (Fortress) digital twin with URDF/Xacro and ros_gz, integrating LiDAR, IMU, and camera sensing; implemented Nav2 and AMCL for autonomous navigation, and **built and debugged the physical platform across CAD, electronics, sensor integration, and hardware-software bringup**.

Contingent Member at Inter IIT Tech Meet 14.0

- Built and integrated an autonomous warehouse inventory robot across **mechanical assembly, electronics debugging, ROS 2, and computer vision**, working across the physical and software stack.
- Developed and optimized a **QR-code detection pipeline** for reliable rack identification across varying viewing angles, tuning approach direction and motion speed to improve detection during autonomous scanning.

PROJECTS

Latent-Space Inference of Dynamic Multi-Agent Systems

Ongoing Research Project under Nishant Ramakuru, Rubotix.ai

- Built a law-agnostic multi-agent physics engine from scratch simulating five interaction laws: **Boids, Leader-Follower, Predator-Prey, Vicsek, and Spring Network.** Generated a large domain-randomized benchmark dataset to encourage learning of underlying dynamics rather than surface-level statistics across diverse simulated environments.
- Implemented and compared **LSTM**, **Transformer**, and **Graph Neural Network (GNN)** architectures with permutation-invariant, mask-aware representations, while developing a reproducible training and evaluation framework featuring seeded experiments, trajectory and classification metrics, and latent-space visualization and analysis.

LLM Dyad | Context-Aware Local AI Assistant

- Built a locally hosted AI assistant integrated with **WhatsApp**, dynamically invoked through natural-language triggers such as *"***Hey Dyad***"*, enabling context-aware participation in conversations using **Ollama** for local LLM inference.
- Engineered an end-to-end **retrieval and inference** pipeline with a directory-based memory architecture separating transient conversation history from **persistent user-controlled memory**, supporting **context retrieval**, **prompt construction**, **response generation**, and **conversation persistence**.

ClauseFinder | RAG over Legal Contracts

- Building a **retrieval-augmented Q&A** system over the **CUAD dataset** (~510 real commercial contracts with expert-labeled clauses across 41 clause categories), deploying a **containerized Qdrant vector store** and local **sentence-transformers** embedding pipeline for private, citation-grounded clause lookup with no data leaving the host.

- Designing a **clause-aware chunking strategy** benchmarked against a fixed-window baseline, with a scored eval set (~40 held-out Q&A pairs) measuring **retrieval hit-rate@k, answer correctness, and citation accuracy**.
- Packaging as a **Dockerized FastAPI service** (alongside **Qdrant**) with a swappable LLM backend (**local Ollama**), deployable to a **self-hosted home-lab server** and exposed privately over **Tailscale** for multi-device access.

Multi-Source Research Agentic Pipeline

- Architected a **five-stage agentic pipeline** in Python that decomposes a research question, searches web sources via **Tavily**, and extracts verifiable claims, chaining **local LLM inference** (Ollama) so each stage **reasons over prior outputs** rather than a single retrieve-and-generate pass.
- Engineered the **cross-verification stage** that clusters claims by topic and flags agreement, contradiction, or single-source coverage, generating cited reports that surface source disagreement instead of silently blending conflicting facts.
- Hardened model output with **JSON-mode schema validation** and **enum coercion**, packaging it reproducibly with **uv** around a dataclass claim schema.

POSITIONS OF RESPONSIBILITY

Core Member, Epoch -- AI/ML Club, IIT Hyderabad

- Led development of a **VLA model** for robotic manipulation using **PyTorch** and **RoboSuite**. Conducted **Deep Learning** sessions and mentored **50**+ students on technical concepts and project development.

Head at LitSoc, IIT Hyderabad

- Led a **20-member team** organizing campus-wide literary and film programming, including discussions, screenings, fairs, competitions, and cross-club collaborations.

SKILLS

Programming : Python, C, C++, JavaScript, SQL, Bash

Frameworks & Tools : Git, Docker, ROS 2, React, FastAPI, Linux, CMake, PostgreSQL, Ollama, uv

Machine Learning : PyTorch, TensorFlow, HuggingFace, NumPy, Scikit-learn, PyTorch Geometric, OpenCV, Gymnasium.

RELEVANT COURSES

Data Structures & Applications, Artificial Intelligence, Linear Algebra, Probability, Advanced Mathematical Physics.